

PatientTriage.ai

Comprehensive Technical System Documentation & Architecture Guide

Project: Accenture Innovation Challenge 2026 — Track 2	Institution: Indian Institute of Technology (IIT) Guwahati
Team: WeWillWin (Abhinav Rai, Karan Aditya, Jai A Mishra)	Deployment Target: Local-First / Non-Device CDS (Zero PHI Outflow)

Executive Summary: PatientTriage.ai is an intelligent clinical decision-support layer engineered to optimize Emergency Department (ED) patient sequencing and mitigate preventable wait-time mortality. Built on two fundamental tenets—Asymmetric Risk Optimization (where False Negatives are severely penalized) and Deterministic Safety Floors (guaranteeing that AI can only escalate priority, never downgrade)—the system bridges demographic-stratified predictive machine learning with deterministic ESI Version 5 safety guidelines and a real-time Continuous Max Heap scheduling engine.

1. Core Principles & Architectural Tenets

Standard ED triage workflows face critical bottlenecks: triage mis-prioritization is a mortality issue, over 50% of patients arrive with zero prior medical history, and unmonitored lower-acuity patients can rapidly decompensate while waiting in static FIFO queues. PatientTriage.ai addresses these operational realities through four core architectural pillars:

- Asymmetric Risk Optimization:** Missing a critical patient (undertriage / False Negative) carries catastrophic clinical consequences. Custom loss penalties during training heavily penalize False Negatives ($\alpha = 18.0$ to 28.0) over False Positives ($\beta = 1.0$).
- Deterministic ESI v5 Safety Floors:** ML predictions are strictly bounded by uncompromisable clinical rules. The final acuity level is determined by $\text{Final_ESI} = \min(\text{ML_ESI_Recommendation}, \text{ABCDE_ESI_Floor})$, ensuring priority is only ever automatically escalated.
- Zero-History Resilience:** Built to maintain high precision even when historical records and comorbidities are completely unverified (50% baseline), routing immediately on presenting vitals and frailty indicators.
- Two-Stage Processing Architecture:** Stage 1 provides a discrete human-in-the-loop classification checkpoint with SHAP explainability and uncertainty scoring; Stage 2 manages real-time queue positioning via a continuous Max Heap scheduler.
- Non-Device CDS Compliance:** Functions strictly as a Non-Device Clinical Decision Support (CDS) tool under FDA/CE guidelines, ensuring transparent reasoning traces and unrestricted clinician override authority.

2. Demographic-Specific Machine Learning Architecture

Deploying a single adult-trained model across all demographics introduces silent safety risks because physiological baselines differ vastly. PatientTriage.ai decouples risk estimation into three targeted XGBoost agents:

Agent Cohort	Target Population	Asymmetric Penalty (α)	Features Evaluated	Performance Metrics
--------------	-------------------	---------------------------------	--------------------	---------------------

Geriatric Agent	Ages 65+ diminished reserve	$\alpha = 23.0$ (FN penalized 23x)	HR, RR, SpO2, SBP, Temp, CFS Frailty Scale (1-9), Prior History, Comorbidity Count	ROC-AUC: 0.728 Catch Rate: 87.5% Threshold: 0.504
Adult Agent	Ages 18–64 acute derangement	$\alpha = 18.0$ (FN penalized 18x)	HR, RR, SpO2, SBP, Temp, Prior History, Comorbidity Count	ROC-AUC: 0.805 Catch Rate: 88.3% Threshold: 0.504
Pediatric Agent	Ages <18 rapid decompensation	$\alpha = 28.0$ (FN penalized 28x)	Age, HR, RR, SpO2, SBP, Temp, Prior History, Comorbidity Count	ROC-AUC: 0.695 Catch Rate: 75.0% Threshold: 0.504

Asymmetric Binary Cross-Entropy Objective Formulation:

The custom XGBoost training objective computes first (grad) and second (hess) derivatives as:

grad = p * (α * y + β * (1 - y)) - α * y

hess = p * (1 - p) * (α * y + β * (1 - y))

where $p = 1 / (1 + \exp(-\text{margin}))$, y is the 30-day mortality/ICU critical outcome label, and $\beta = 1.0$.

3. Stage 1: Deterministic ESI v5 Safety Floor Engine

To prevent 'normal-vitals danger' cases from slipping through probabilistic models, the system incorporates an uncompromisable pure Python rules engine compliant with the ESI Version 5 handbook:

Decision Point	Clinical Evaluation Logic	Acuity Floor
Decision Point A (decision_a.py)	Immediate life-saving interventions required, apneic/pulseless, severe agonal respiratory distress, acute unresponsiveness (AVPU='U' or GCS ≤ 8).	ESI Level 1 (Hard Locked)
Decision Point B (decision_b.py)	High-risk situation, acute altered mental status (AVPU in ['V','P'] or GCS < 14), severe pain/distress, acute chest pain/STEMI suspect, acute stroke symptoms (FAST+), anaphylaxis.	ESI Level 2 (High Priority)
Decision Point D (decision_d.py)	Age-Stratified Danger Zone Vitals: - Infant (<1y): HR > 160, RR > 45, SpO2 < 92% - Toddler (1-4y): HR > 140, RR > 35, SpO2 < 92% - Adolescent (5-17y): HR > 110, RR > 24, SpO2 < 92% - Adult (18-64y): HR > 100, RR > 20, SpO2 < 92% Geriatric Frailty Guard: Age ≥ 65, CFS ≥ 5, and HR > 90 bpm.	ESI Level 2 (Vital Escalation)
Decision Point C (decision_c.py)	Resource estimation based on labs, imaging, IV medications, and procedures. 2+ Resources → ESI 3 1 Resource → ESI 4 0 Resources → ESI 5. Zero-History/Arrival Fallback: Stubs uncollected intake as insufficient_data (defaults to standard ESI 3).	ESI Level 3/4/5 (Standard Care)

4. Explainability, Epistemic Uncertainty & Grounding

To avoid black-box obscurity and eliminate LLM hallucinations, PatientTriage.ai implements a three-tier verification layer:

- Tree-Level Ensemble Variance Score (uncertainty.py):** Evaluates prediction consensus across sequential boosting sub-ranges of trees (e.g. rounds 15, 30, 45... 150). High standard deviation across iterations indicates conflicting feature signals. The score maps to a 0–100% confidence metric. If confidence drops below **20.0%**, the system automatically triggers a *Mandatory Human Review Gate*.
- SHAP Explainer Service (shap_explainer.py):** Computes exact Shapley attributions using shap.TreeExplainer on the active demographic model. Features are categorized into risk-increasing vs. protective drivers and paired with age-banded clinical reference ranges from data/esi_reference_tables.json.

- **Grounding Validator (grounding.py):** Enforces structured JSON verification before rendering explanations. Rejects narrative hallucinations (e.g. claiming a normal SpO2 of 98% elevated risk when SHAP is negative, or citing triggered rules not generated by the ESI engine).

5. Stage 2: Continuous Max Heap Dynamic Priority Scheduler

Static FIFO waitlists fail in ED triage because unmonitored lower-acuity patients deteriorate over time. Stage 2 implements a continuous Max Heap priority queue where sorting scores update dynamically:

Continuous Dynamic Priority Scoring Equation:

$$\text{Priority Score}(t) = W_{\text{floor}} \times (6 - \text{ESI}_{\text{final}}) + W_{\text{risk}} \times P_{\text{risk}} + W_{\text{time}} \times \ln(1 + t_{\text{wait}} / \tau)$$

- **Acuity Base ($W_{\text{floor}} = 1000$):** Inverts ESI level (ESI 1=5000, ESI 2=4000, ESI 3=3000, ESI 4=2000, ESI 5=1000). Provides strict mathematical guarantees that an ESI 2 patient will ALWAYS outrank an ESI 4 patient regardless of waiting time (no cross-tier inversion).
- **Intra-Tier Sorting ($W_{\text{risk}} = 100$):** Uses continuous P_{risk} (0.0 to 1.0) to sort patients within the same ESI tier.
- **Sub-Linear Time Decay ($W_{\text{time}} = 15$, $\tau = 30$):** Applies logarithmic multiplier to wait time in minutes to prevent queue starvation without breaching clinical safety tiers.
- **Surge Volume Mitigation:** When active queue length exceeds threshold (≥ 15 patients) or during surge shifts, W_{time} scales to **30.0**, accelerating throughput for waiting patients.
- **Deterioration Re-Sort & Overrides:** If patient vitals worsen or a nurse enters an override, the heap re-indexes in $O(\log N)$ time instantly.

6. Agent Orchestration Loop & Local LLM Integration

PatientTriage.ai connects predictive ML, deterministic safety floors, and explainability via an agent orchestration loop (agent/orchestrator.py):

- **Local LLM Target:** Integrates with local gemma4:12b via Ollama (/api/chat) using tool-calling with "think": false to optimize VRAM utilization on local hardware (~8.5GB VRAM target). Zero PHI leaves the device.
- **Exposed Clinical Tools:** predict_esi_xgboost, get_shap_explanation, run_esi_rule_engine, and lookup_reference_range.
- **Dual-Mode Standalone Fallback:** If Ollama is offline or not installed, the system automatically routes through an internal deterministic clinical reasoning synthesizer with 0 ms connection overhead, ensuring 100% offline uptime and zero setup barriers.

7. Repository Structure & Verification Results

```
patienttriage/
■■■ models/ [geriatric_xgb.json, adult_xgb.json, pediatric_xgb.json, models_metadata.json]
■■■ data/ [esi_reference_tables.json]
■■■ rule_engine/ [decision_a.py, decision_b.py, decision_c.py, decision_d.py, engine.py]
■■■ scheduler/ [scoring.py, max_heap.py]
■■■ explain/ [shap_explainer.py, uncertainty.py, grounding.py]
■■■ agent/ [tools.py, orchestrator.py, llm_client.py]
■■■ api/ [main.py]
■■■ ui/ [app.py]
■■■ scripts/ [generate_data.py, train_models.py]
■■■ tests/ [test_rule_engine.py, test_scheduler.py, test_grounding.py, test_api.py]
```

Automated Test Suite Verification Matrix (19/19 Passed):

Test Module	Scope & Validation Scenarios	Result
test_rule_engine.py	Decision Points A (Arrest), B (Stroke/AMS), C (Resource Fallback), D (Ped & Adult Danger Vitals), Frailty Guard, min(ML, Floor) bounding.	PASSED (7/7)
test_scheduler.py	Dynamic Priority formula, strict tier preservation (ESI 2 > ESI 4), intra-tier risk sorting, surge acceleration, heap re-indexing on override.	PASSED (5/5)
test_grounding.py	Grounding validation, hallucination rejection on inverted SHAP claims, acuity floor breach rejection, numerical value fidelity checks.	PASSED (4/4)
test_api.py	FastAPI endpoints: /health, /analyze-patient, /queue, /queue/add, /override, /queue/pop-next.	PASSED (3/3)

8. Operational Run Guide & Execution Steps

Step / Target	Command Line	Operational Function
1. Launch Interactive Streamlit UI	python main.py --ui	Starts the full web dashboard at http://localhost:8501 with live intake presets, SHAP waterfall charts, real-time waiting room queue, surge mode controls, and clinician override modals.
2. Launch FastAPI REST Backend	python main.py --api	Starts the high-performance Uvicorn backend at http://localhost:8000. Swagger API documentation is accessible at http://localhost:8000/docs.
3. Execute Full Test Suite	python -m pytest patienttriage/tests -v	Executes all 19 unit and integration tests covering the rules engine, scheduler, grounding validator, and REST endpoints.
4. Run Terminal Batch Simulation	python main.py --simulate	Generates synthetic ED cohorts and evaluates agreement states, ML risk escalation, and deterministic safety floor triggers in terminal format.
5. Re-train Demographic Models	python -m patienttriage.scripts.train_models	Re-synthesizes training data and serializes geriatric_xgb.json, adult_xgb.json, and pediatric_xgb.json.

CDS Regulatory Notice: PatientTriage.ai is designed in accordance with Non-Device Clinical Decision Support (CDS) governance. It generates transparent, grounded clinical insights and safety boundaries. Attending healthcare professionals retain full decision autonomy and override authority.